

Information Field Theory

Economics and Complex Systems

IFT-Econ v1.0

Juan Diego Vicente Gabancho

April 2026

The application of Information Field Theory (IFT) to economics and complex systems is presented. Financial markets are modeled as informational fields $\rho(x, t)$ where x represents the agents' strategy space. Prices emerge as informational gradients $P \sim |\nabla \ln \rho|$, reflecting the intensity of competition among strategies. The wealth distribution follows a Pareto law as a consequence of maximizing informational entropy under constraints. Financial crashes are interpreted as collapses of informational coherence: when $\tau\Xi < K_{\text{market}}$, the system loses temporal coherence and transitions to a disordered state. Historical events (the 2010 flash crash, the 2008 crisis) are analyzed under this framework, and early-warning indicators based on Ξ and τ are proposed.

Contents

1	Introduction: Economics as an Informational System	4
1.1	Motivation	4
1.2	Central Thesis	4
2	Markets as Informational Fields	5
2.1	Strategy Space	5
2.2	Informational Flow Dynamics	5
3	Prices as Informational Gradients	6
3.1	Informational Equilibrium Price	6
3.2	Volatility as Gradient Variance	6
3.3	Market Index	6
4	Wealth Distribution as Entropy Maximization	7
4.1	Maximum Entropy Principle	7
4.2	Derivation from IFT	7
5	Crashes as Coherence Collapses	8
5.1	Market Stability Criterion	8
5.2	Crash Mechanism	8
5.3	Early-Warning Indicator	8
6	Analysis of Historical Events	9
6.1	Flash Crash of May 6, 2010	9
6.2	Financial Crisis of 2008	9
7	Predictions and Applications	10
7.1	IFT Indicators for Financial Regulation	10
7.2	Testable Predictions	10
8	Master Formula of IFT-Econ	11
9	Final Declaration	12
9.1	Affirmations	12
9.2	Open Problems	12

Chapter 1

Introduction: Economics as an Informational System

1.1 Motivation

Neoclassical economics assumes rational agents, perfect information, and equilibrium. Reality shows agents with bounded rationality, asymmetric information, and out-of-equilibrium dynamics with sudden crashes. IFT provides a natural framework for modeling these features: the informational field $\rho(x, t)$ captures the distribution of agents' strategies, beliefs, and resources.

1.2 Central Thesis

Theorem 1.1 (IFT-Econ Thesis). *Financial markets are informational systems where the field $\rho(x, t)$ represents the density of agents adopting strategy x at time t . Prices, volatility, and crashes emerge from the geometry and dynamics of ρ , not from equilibrium assumptions.*

Chapter 2

Markets as Informational Fields

2.1 Strategy Space

Each agent i adopts a strategy $x_i \in \Omega$ (a vector in a d -dimensional space that may include: asset allocation, leverage, time horizon, risk tolerance). The informational field is defined as:

$$\rho(x, t) = \frac{1}{N} \sum_{i=1}^N \delta(x - x_i(t)), \quad (2.1)$$

or, in the continuous limit, as a smooth density.

2.2 Informational Flow Dynamics

Agents adjust their strategies following the gradient of a collective utility function:

$$\partial_t \rho = \nabla \cdot \left(D \rho \nabla \frac{\delta \mathcal{F}}{\delta \rho} \right), \quad (2.2)$$

where $\mathcal{F}[\rho] = \mathcal{U}[\rho] - T_{\text{market}} S[\rho]$ is a market free energy, $\mathcal{U}[\rho]$ is the collective expected utility, and T_{market} is a “market temperature” measuring uncertainty.

Chapter 3

Prices as Informational Gradients

3.1 Informational Equilibrium Price

The price of an asset emerges as the informational gradient in strategy space:

$$\boxed{P(x, t) \propto |\nabla \ln \rho(x, t)| = \Xi(x, t)}. \quad (3.1)$$

Interpretation: in regions of strategy space where ρ varies rapidly, there is high competition and prices adjust quickly. In homogeneous regions ($\Xi \approx 0$), prices are stable.

3.2 Volatility as Gradient Variance

$$\sigma^2(t) = \text{Var}_x[\Xi(x, t)] = \int \rho(x, t) (\Xi(x, t) - \bar{\Xi}(t))^2 dx. \quad (3.2)$$

Market volatility is the dispersion of informational gradients among strategies.

3.3 Market Index

A market index (e.g., S&P 500) corresponds to the average informational gradient:

$$I(t) \propto \bar{\Xi}(t) = \int \rho(x, t) \Xi(x, t) dx. \quad (3.3)$$

Chapter 4

Wealth Distribution as Entropy Maximization

4.1 Maximum Entropy Principle

The wealth distribution observed in real economies follows a Pareto law (power law) for the upper tails. IFT derives this from the maximum entropy principle:

$$\rho(w) \propto w^{-\alpha-1} e^{-w/w_c}, \quad (4.1)$$

where $\alpha \approx 1.5 - 2$ (Pareto index) and w_c is a cutoff scale.

4.2 Derivation from IFT

Maximizing $S[\rho] = - \int \rho(w) \ln \rho(w) dw$ subject to:

1. Normalization: $\int \rho(w) dw = 1$,
2. Fixed mean wealth: $\int w \rho(w) dw = \bar{w}$,
3. Logarithmic (informational) constraint: $\int \ln w \rho(w) dw = \overline{\ln w}$ (constant),

produces exactly the gamma distribution, whose tail is a Pareto law.

Chapter 5

Crashes as Coherence Collapses

5.1 Market Stability Criterion

A market is stable if it maintains informational coherence:

$$\tau_{\text{market}} \cdot \Xi_{\text{market}} > K_{\text{market}}, \quad (5.1)$$

where τ_{market} is the coherence time of strategies (how long agents maintain their positions) and Ξ_{market} is the average informational gradient.

5.2 Crash Mechanism

When $\tau\Xi < K_{\text{market}}$:

1. Agents change strategies faster than the market can absorb (high frequency).
2. The informational gradient collapses: all strategies converge to a single point (mass sell-off).
3. Entropy drops abruptly ($dS/dt < 0$ locally, violation of the H-theorem).
4. The system transitions to a new equilibrium with lower ρ (lower total wealth).

5.3 Early-Warning Indicator

The product $\tau(t) \cdot \Xi(t)$ measured in real time can serve as an early-warning indicator. A sustained drop below a critical threshold K_{market} precedes a crash.

Chapter 6

Analysis of Historical Events

6.1 Flash Crash of May 6, 2010

- τ collapsed to milliseconds due to high-frequency algorithmic trading.
- Ξ spiked locally (mass sell-off concentrated in few strategies).
- $\tau\Xi < K_{\text{market}}$: coherence collapse.
- Recovery in minutes: the system found a new equilibrium with redistributed ρ .

6.2 Financial Crisis of 2008

- Ξ gradually decreased over years (convergence of strategies toward mortgage instruments).
- τ was artificially lengthened (agents maintained positions without questioning them).
- The collapse occurred when Ξ fell below a critical threshold (everyone wanted to sell at the same time).

Chapter 7

Predictions and Applications

7.1 IFT Indicators for Financial Regulation

1. **Coherence Index:** $\tau(t) \cdot \Xi(t)$ measured in real time for each sector.
2. **Informational Diversity:** $S[\rho]$ (entropy) as a measure of strategy diversity.
3. **Risk Concentration:** $\text{Var}[\Xi]$ as a measure of systemic risk.

7.2 Testable Predictions

1. Crashes are preceded by a drop in $\tau\Xi$ below a historical threshold.
2. The distribution of financial returns should follow an asymmetric Laplace distribution (consistent with the informational flow model).
3. Volatility should cluster temporally (volatility clustering) as a consequence of the dynamics of Ξ .

Chapter 8

Master Formula of IFT-Econ

$$\rho(x, t) = \text{density of agents in strategy } x, \quad (8.1)$$

$$\partial_t \rho = \nabla \cdot \left(D \rho \nabla \frac{\delta \mathcal{F}}{\delta \rho} \right) \quad (\text{Strategy flow}), \quad (8.2)$$

$$P \propto \Xi = |\nabla \ln \rho| \quad (\text{Price as gradient}), \quad (8.3)$$

$$\text{Crash: } \tau \Xi < K_{\text{market}} \quad (\text{Coherence collapse}), \quad (8.4)$$

$$S[\rho] = - \int \rho \ln \rho \, dx \quad (\text{Market diversity}). \quad (8.5)$$

Chapter 9

Final Declaration

IFT-Econ demonstrates that financial markets can be modeled as informational systems where prices emerge from gradients, wealth follows maximum-entropy distributions, and crashes are coherence collapses. The framework provides quantitative early-warning indicators and unifies observed phenomena (Pareto law, volatility clustering, sudden crashes) under a single formalism.

$$\boxed{\text{Market} = \text{Informational field of strategies}}.$$

$$\boxed{\text{Price} = \text{Informational gradient } \Xi}.$$

$$\boxed{\text{Crash} = \text{Coherence collapse } \tau\Xi < K_{\text{market}}}.$$

9.1 Affirmations

1. Markets are out-of-equilibrium informational systems.
2. Prices emerge as informational gradients.
3. Wealth distribution follows the maximum-entropy principle.
4. Crashes are informational phase transitions.
5. The $\tau\Xi$ indicator provides early warning of instability.

9.2 Open Problems

1. Estimate $\tau(t)$ and $\Xi(t)$ in real time from market data.
2. Calibrate K_{market} for different markets and time scales.
3. Validate the asymmetric Laplace prediction for financial returns.
4. Develop agent-based simulation models that implement IFT dynamics.
5. Extend the framework to cryptoasset markets and decentralized finance.